KUBERNETES

Complete Commands Reference Guide

kubectl • Cluster • Pods • Services • Deployments • Troubleshooting

1. CLUSTER SETUP & INFO

kubectl Version & Config

Command	Description
`kubectl version`	Show client and server version
`kubectl version --client`	Show only client version
`kubectl cluster-info`	Show cluster endpoint info
`kubectl cluster-info dump`	Full cluster diagnostic dump
`kubectl config view`	Show kubeconfig file
`kubectl config get-contexts`	List all contexts
`kubectl config current-context`	Show current active context
`kubectl config use-context ollama-agent`	Switch to a different context
`kubectl config set-context --current --namespace=ollama-agent`	Set default namespace for current context
`kubectl api-resources`	List all API resource types
`kubectl api-versions`	List all API versions
`kubectl get nodes`	List all nodes in cluster
`kubectl get nodes -o wide`	List nodes with IP and OS info
`kubectl describe node ollama-agent-control-plane`	Detailed info about a node
`kubectl top nodes`	Show CPU/memory usage of nodes
`kubectl cordon <node>`	Mark node as unschedulable
`kubectl uncordon <node>`	Mark node as schedulable again
`kubectl drain <node> --ignore-daemonsets`	Evict all pods from a node

KIND Cluster Commands

Command	Description
`kind create cluster`	Create default KIND cluster
`kind create cluster --name ollama-agent`	Create named KIND cluster
`kind create cluster --name ollama-agent --config <file.yaml>`	Create with custom config
`kind get clusters`	List all KIND clusters
`kind delete cluster --name ollama-agent`	Delete a KIND cluster

Command	Description
`kind load docker-image <image> --name <cluster>`	Load local Docker image into KIND
`kind export kubeconfig --name ollama-agent`	Export kubeconfig for KIND cluster

2. NAMESPACES

Command	Description
`kubectl get namespaces`	List all namespaces
`kubectl get ns`	Short form — list namespaces
`kubectl create namespace ollama-agent`	Create a namespace
`kubectl delete namespace ollama-agent`	Delete namespace and ALL resources inside
`kubectl describe namespace ollama-agent`	Show namespace details and resource quotas
`kubectl get all -n ollama-agent`	List all resources in a namespace
`kubectl get all --all-namespaces`	List all resources across all namespaces
`kubectl get all -A`	Short form — all resources, all namespaces
`kubectl apply -f namespace.yaml`	Create namespace from YAML file
`-n ollama-agent`	Flag: target a specific namespace (used with any command)
`--namespace=ollama-agent`	Long form of -n flag
`--all-namespaces or -A`	Target all namespaces

[TIP] Always use -n ollama-agent with every kubectl command to avoid hitting the wrong namespace.

3. PODS

Get & Describe Pods

Command	Description
`kubectl get pods`	List pods in default namespace
`kubectl get pods -n ollama-agent`	List pods in specific namespace
`kubectl get pods -A`	List pods in all namespaces
`kubectl get pods -o wide`	List pods with node, IP, and more
`kubectl get pods -w`	Watch pods in real time
`kubectl get pods --show-labels`	Show pod labels
`kubectl get pods -l app=ollama-agent`	Filter pods by label
`kubectl describe pod ollama-backend-xxxxx -n ollama-agent`	Full details: events, status, containers
`kubectl get pod ollama-backend-xxxxx -o yaml`	Get pod spec as YAML
`kubectl get pod ollama-backend-xxxxx -o json`	Get pod spec as JSON

Pod Logs

Command	Description
`kubectl logs ollama-backend-xxxxx -n ollama-agent`	Get pod logs
`kubectl logs -f ollama-backend-xxxxx -n ollama-agent`	Stream logs in real time (-f = follow)
`kubectl logs ollama-backend-xxxxx -c <container> -n ollama-agent`	Logs from specific container in pod
`kubectl logs ollama-backend-xxxxx --previous -n ollama-agent`	Logs from crashed/previous container
`kubectl logs ollama-backend-xxxxx --tail=100 -n ollama-agent`	Last 100 lines of logs
`kubectl logs ollama-backend-xxxxx --since=1h -n ollama-agent`	Logs from last 1 hour
`kubectl logs -f deployment/ollama-agent -n ollama-agent`	Stream logs from a deployment
`kubectl logs -f statefulset/ollama-agent -n ollama-agent`	Stream logs from a statefulset

Pod Exec & Shell

Command	Description
`kubectl exec -it <pod> -n ollama-agent -- /bin/sh`	Open shell in pod (Alpine/slim images)
`kubectl exec -it <pod> -n ollama-agent -- /bin/bash`	Open bash shell in pod
`kubectl exec -it <pod> -c <container> -n ollama-agent -- /bin/sh`	Shell in specific container
`kubectl exec <pod> -n ollama-agent -- ls /app`	Run single command in pod
`kubectl exec <pod> -n ollama-agent -- env`	Print environment variables in pod
`kubectl exec <pod> -n ollama-agent -- cat /etc/hosts`	Print hosts file in pod
`kubectl exec -it statefulset/ollama-agent -n ollama-agent -- mongosh`	Open mongosh in MongoDB StatefulSet

Pod Lifecycle

Command	Description
`kubectl run ollama-agent --image=<image>`	Create a standalone pod
`kubectl run ollama-agent --image=<image> --rm -it -- /bin/sh`	Temp debug pod — auto-deletes on exit
`kubectl delete pod ollama-backend-xxxxx -n ollama-agent`	Delete a pod (recreated if in Deployment)
`kubectl delete pod ollama-backend-xxxxx -n ollama-agent --force`	Force delete stuck pod
`kubectl delete pod ollama-backend-xxxxx -n ollama-agent --grace-period=0`	Immediate deletion (use carefully)
`kubectl get pod ollama-backend-xxxxx -n ollama-agent -o jsonpath='{.status.phase}'`	Get pod status programmatically

4. DEPLOYMENTS

Create & Manage Deployments

Command	Description
`kubectl get deployments -n ollama-agent`	List all deployments
`kubectl get deploy -n ollama-agent`	Short form
`kubectl describe deployment ollama-agent -n ollama-agent`	Full deployment details
`kubectl apply -f deployment.yaml`	Create/update deployment from file
`kubectl delete deployment ollama-agent -n ollama-agent`	Delete a deployment
`kubectl create deployment ollama-agent --image=<image>`	Create deployment imperatively
`kubectl get deployment ollama-agent -o yaml -n ollama-agent`	Export deployment as YAML

Scaling & Rolling Updates

Command	Description
`kubectl scale deployment ollama-agent --replicas=3 -n ollama-agent`	Scale deployment to 3 replicas
`kubectl rollout status deployment/ollama-agent -n ollama-agent`	Watch rollout progress
`kubectl rollout history deployment/ollama-agent -n ollama-agent`	View rollout history
`kubectl rollout undo deployment/ollama-agent -n ollama-agent`	Rollback to previous version
`kubectl rollout undo deployment/ollama-agent --to-revision=2 -n ollama-agent`	Rollback to specific revision
`kubectl rollout restart deployment/ollama-agent -n ollama-agent`	Force restart all pods (rolling)
`kubectl rollout pause deployment/ollama-agent -n ollama-agent`	Pause a rolling update
`kubectl rollout resume deployment/ollama-agent -n ollama-agent`	Resume a paused rollout
`kubectl set image deployment/ollama-agent <container>=<image>:<tag> -n ollama-agent`	Update image in deployment

> *[OK] Always use kubectl rollout status after kubectl apply to confirm the rollout completed successfully.*

5. SERVICES (SVC)

Get & Describe Services

Command	Description
`kubectl get services -n ollama-agent`	List all services

Command	Description
`kubectl get svc -n ollama-agent`	Short form
`kubectl get svc -A`	All services across all namespaces
`kubectl get svc -o wide -n ollama-agent`	Services with selector and ports
`kubectl describe svc ollama-agent -n ollama-agent`	Full service details with endpoints
`kubectl get endpoints -n ollama-agent`	List endpoints (pod IPs behind services)
`kubectl get ep ollama-agent -n ollama-agent`	Endpoints for specific service
`kubectl apply -f service.yaml`	Create/update service from file
`kubectl delete svc ollama-agent -n ollama-agent`	Delete a service
`kubectl get svc ollama-agent -o yaml -n ollama-agent`	Export service as YAML

Service Types Explained

Command	Description
`ClusterIP (default)`	Internal only — accessible inside cluster. No external access.
`NodePort`	Opens port (30000–32767) on every node. External access via NodeIP:NodePort.
`LoadBalancer`	Cloud load balancer (AWS ELB). Not available in KIND — use port-forward.
`ExternalName`	Maps service to external DNS name. No proxying.
`Headless (clusterIP: None)`	No load balancing. DNS returns individual pod IPs. Used by StatefulSets.

Port Forwarding

Command	Description
`kubectl port-forward svc/ollama-agent 8080:80 -n ollama-agent`	Forward local 8080 to service port 80
`kubectl port-forward pod/ollama-agent 8080:80 -n ollama-agent`	Forward to specific pod
`kubectl port-forward svc/ollama-agent 8080:80 -n ollama-agent --address 0.0.0.0`	Expose on all interfaces (EC2 access)
`kubectl port-forward svc/ollama-agent 8080:80 -n ollama-agent &`	Run port-forward in background
`kubectl port-forward -n ingress-nginx svc/ingress-nginx-controller 8080:80`	Forward Nginx Ingress controller

6. STATEFULSETS

Command	Description
`kubectl get statefulsets -n ollama-agent`	List all StatefulSets
`kubectl get sts -n ollama-agent`	Short form

Command	Description
`kubectl describe sts ollama-agent -n ollama-agent`	Detailed StatefulSet info
`kubectl apply -f statefulset.yaml`	Create/update StatefulSet from file
`kubectl delete sts ollama-agent -n ollama-agent`	Delete StatefulSet (pods also deleted)
`kubectl scale sts ollama-agent --replicas=3 -n ollama-agent`	Scale StatefulSet
`kubectl rollout status sts/ollama-agent -n ollama-agent`	Watch StatefulSet rollout
`kubectl rollout restart sts/ollama-agent -n ollama-agent`	Restart StatefulSet pods
`kubectl exec -it ollama-mongodb-0 -n ollama-agent -- /bin/sh`	Shell into first StatefulSet pod
`kubectl logs ollama-mongodb-0 -n ollama-agent`	Logs from first StatefulSet pod

[TIP] StatefulSet pods are named with stable index: mongodb-0, mongodb-1. Always start from index 0.

7. CONFIGMAPS & SECRETS

ConfigMaps

Command	Description
`kubectl get configmaps -n ollama-agent`	List ConfigMaps
`kubectl get cm -n ollama-agent`	Short form
`kubectl describe cm ollama-agent -n ollama-agent`	Show ConfigMap contents
`kubectl apply -f configmap.yaml`	Create/update ConfigMap from file
`kubectl delete cm ollama-agent -n ollama-agent`	Delete a ConfigMap
`kubectl create configmap ollama-agent --from-literal=KEY=VALUE -n ollama-agent`	Create ConfigMap from literal value
`kubectl create configmap ollama-agent --from-file=<file> -n ollama-agent`	Create ConfigMap from file
`kubectl get cm ollama-agent -o yaml -n ollama-agent`	Export ConfigMap as YAML

Secrets

Command	Description
`kubectl get secrets -n ollama-agent`	List all secrets
`kubectl describe secret ollama-agent -n ollama-agent`	Show secret metadata (not values)
`kubectl apply -f secret.yaml`	Create/update secret from file
`kubectl delete secret ollama-agent -n ollama-agent`	Delete a secret
`kubectl get secret ollama-agent -o yaml -n ollama-agent`	Export secret (values are base64)

Command	Description
`kubectl get secret ollama-agent -o jsonpath='{.data.KEY}' -n ollama-agent | base64 --decode`	Decode a secret value
`kubectl create secret generic ollama-agent --from-literal=KEY=VALUE -n ollama-agent`	Create secret imperatively
`echo -n 'value' | base64`	Encode a value for use in secret.yaml
`echo 'encoded==' | base64 --decode`	Decode a base64 secret value

[WARN] Never commit secret.yaml with real values to Git. Always use .gitignore or external secret managers.

8. PERSISTENT VOLUMES & STORAGE

Command	Description
`kubectl get pvc -n ollama-agent`	List PersistentVolumeClaims
`kubectl get pv`	List PersistentVolumes (cluster-wide)
`kubectl describe pvc ollama-agent -n ollama-agent`	Full PVC details including status
`kubectl describe pv ollama-agent`	Full PV details
`kubectl get storageclass`	List storage classes
`kubectl describe storageclass ollama-agent`	Storage class details
`kubectl apply -f pvc.yaml`	Create PVC from file
`kubectl delete pvc ollama-agent -n ollama-agent`	Delete PVC (data may be lost!)
`kubectl get pvc -n ollama-agent -o wide`	PVC with storage class and capacity

[WARN] Deleting a PVC deletes the data inside it permanently. Always back up before deleting.

9. INGRESS

Command	Description
`kubectl get ingress -n ollama-agent`	List all Ingress resources
`kubectl get ing -n ollama-agent`	Short form
`kubectl get ingress -A`	Ingress across all namespaces
`kubectl describe ingress ollama-agent -n ollama-agent`	Full Ingress details with rules
`kubectl apply -f ingress.yaml`	Create/update Ingress from file
`kubectl delete ingress ollama-agent -n ollama-agent`	Delete an Ingress
`kubectl get pods -n ingress-nginx`	Check Nginx Ingress Controller pods
`kubectl logs -n ingress-nginx deploy/ingress-nginx-controller`	Ingress controller logs
`kubectl get svc -n ingress-nginx`	Check Ingress controller service

10. HPA — HORIZONTAL POD AUTOSCALER

Command	Description
`kubectl get hpa -n ollama-agent`	List all HPAs
`kubectl describe hpa ollama-agent -n ollama-agent`	HPA details: current vs target metrics
`kubectl apply -f hpa.yaml`	Create/update HPA from file
`kubectl delete hpa ollama-agent -n ollama-agent`	Delete an HPA
`kubectl autoscale deployment ollama-agent --min=2 --max=10 --cpu-percent=70 -n ollama-agent`	Create HPA imperatively
`kubectl top pods -n ollama-agent`	Show CPU/memory usage per pod (needs Metrics Server)
`kubectl top nodes`	Show CPU/memory usage per node

[TIP] HPA requires Metrics Server to be installed. Without it, HPA shows <unknown> for current metrics.

11. APPLY, DELETE & DRY RUN

Command	Description
`kubectl apply -f <file.yaml>`	Create or update resource from file
`kubectl apply -f <directory>/`	Apply all YAML files in a directory
`kubectl apply -f <file.yaml> --dry-run=client`	Preview changes without applying
`kubectl apply -f <file.yaml> --dry-run=server`	Server-side dry run (validates with API)
`kubectl diff -f <file.yaml>`	Show diff between live state and file
`kubectl delete -f <file.yaml>`	Delete resource defined in file
`kubectl delete -f <directory>/`	Delete all resources in directory
`kubectl delete pod,svc,deploy --all -n ollama-agent`	Delete all pods, services, deployments in namespace
`kubectl create -f <file.yaml>`	Create only (fails if already exists)
`kubectl replace -f <file.yaml>`	Replace existing resource (must already exist)
`kubectl patch deployment ollama-agent -p '{...}' -n ollama-agent`	Patch a resource with JSON
`kubectl edit deployment ollama-agent -n ollama-agent`	Edit resource in-place with editor

12. LABELS & SELECTORS

Command	Description
`kubectl get pods --show-labels -n ollama-agent`	Show all labels on pods
`kubectl get pods -l app=backend -n ollama-agent`	Filter pods by label
`kubectl get pods -l app=backend,env=prod -n ollama-agent`	Filter by multiple labels

Command	Description
`kubectl label pod ollama-backend-xxxxx env=prod -n ollama-agent`	Add label to a pod
`kubectl label pod ollama-backend-xxxxx env- -n ollama-agent`	Remove label from a pod (note the dash)
`kubectl get all -l app=ollama-agent -n ollama-agent`	Get all resources with a label
`kubectl get pods -l 'app in (frontend,backend)' -n ollama-agent`	Filter by label set

13. EVENTS & MONITORING

Command	Description
`kubectl get events -n ollama-agent`	List all events in namespace
`kubectl get events -n ollama-agent --sort-by='.lastTimestamp'`	Events sorted by time (most recent last)
`kubectl get events -n ollama-agent --sort-by='.metadata.creationTimestamp'`	Events sorted by creation time
`kubectl get events -A`	Events across all namespaces
`kubectl get events -n ollama-agent --field-selector reason=Failed`	Filter only Failed events
`kubectl top pods -n ollama-agent`	CPU/memory per pod
`kubectl top pods -n ollama-agent --containers`	CPU/memory per container
`kubectl top nodes`	CPU/memory per node

14. TROUBLESHOOTING — A TO Z

Pod Not Starting / CrashLoopBackOff

> *[WARN] CrashLoopBackOff means the container keeps crashing. The --previous flag shows logs before the crash.*

ImagePullBackOff / ErrImagePull

> *[OK] Causes: wrong image name, wrong tag, private registry with no imagePullSecret, typo in image name.*

Pod Stuck in Pending

Service Not Routing Traffic

Ingress Not Working

PVC Stuck in Pending

HPA Showing <unknown> Metrics

StatefulSet Pod Not Starting

Node Not Ready

Force Delete Stuck Resources

> *[WARN] Force deleting pods and removing finalizers can cause data inconsistency. Use only when necessary.*

15. FULL CLUSTER STATUS — QUICK CHECKS

One-liner full cluster status

16. OUTPUT FORMATS & JSONPATH

Command	Description
`kubectl get pods -o yaml -n ollama-agent`	Output as YAML
`kubectl get pods -o json -n ollama-agent`	Output as JSON
`kubectl get pods -o wide -n ollama-agent`	Wide output with extra columns
`kubectl get pods -o name -n ollama-agent`	Output only resource names
`kubectl get pod ollama-agent -o jsonpath='{.status.podIP}' -n ollama-agent`	Get pod IP address
`kubectl get pod ollama-agent -o jsonpath='{.spec.containers[*].name}' -n ollama-agent`	Get all container names
`kubectl get nodes -o jsonpath='{.items[*].metadata.name}'`	Get all node names
`kubectl get pods -n ollama-agent -l app=backend -o jsonpath='{.items[0].metadata.name}'`	Get first matching pod name
`kubectl get svc -n ollama-agent -o jsonpath='{.items[*].spec.clusterIP}'`	Get all ClusterIPs